DP Graded Quiz M5

Question 1

(**Note:** Use Python to solve this question.) Use the **yfinance library** to extract daily stock price data for the firm Exxon Mobil (Ticker: XOM) in the period January 1st, 2021, to December 31st, 2021 (available days in the interval). Use Adjusted Close prices to calculate returns. What is the median continuously compounded daily return for that stock in the period?

- 17.95%
- 17.93%
- 0.1795%
- 0.1793%

Question 2

For which of the following would you need to use Cholesky decomposition?

- To price a call option on a stock
- To calculate Value-at-Risk (VaR) for a portfolio of stocks using simulated data
- To infer the statistical distribution of historical returns
- To price a call option on a bond

Question 3

Given Cholesky decomposition $A = LL^T$ and defining matrix A as:

$$\begin{bmatrix} 0.063 & -0.025 \\ -0.025 & 0.032 \end{bmatrix}$$

which of the following corresponds to L?

- $\begin{bmatrix} 0.063 & 0.05 \\ 0 & 0.032 \end{bmatrix}$

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- $\begin{bmatrix} 0.250 & 0 \\ -0.099 & 0.148 \end{bmatrix}$
- $\begin{bmatrix} 0.250 & -0.099 \\ 0 & 0.148 \end{bmatrix}$
- $\begin{bmatrix} 0.063 & 0 \\ -0.025 & 0.032 \end{bmatrix}$

Question 4

We want to test whether the sample of daily returns we are working with has a mean statistically different from zero. Which of the following would you use?

- Check whether kurtosis equals 3
- Jarque-Bera
- One-sample T-test
- Two-sample T-test

Question 5

Suppose you are investing in a hedge fund that delivers an annual average return after fees of 22%. You plan to invest initially $11,000 and leave the proceeds invested there for the next 30 years. How much will your investment be worth by the end of the period?

- $389,757
- $428,733
- $4.28 million
- $3.89 million

Question 6

(**Note:** Use Python to solve this question.) Use the **yfinance library** in Python to download daily adjusted closed price data for the tickers AAPL, AMZN, and TSLA. Use the period spanning Jan. 1st to July 31st of 2021. Calculate the correlation matrix of their simple daily returns. What does it look like?

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- $\begin{bmatrix} 1 & 0 & 0 \\ 0.591 & 0.8063 & 0 \\ 0.531 & 0.1286 & 0.8370 \end{bmatrix}$
- $\begin{bmatrix} 0.000285 & 0.000160 & 0.000328 \\ 0.000160 & 0.000258 & 0.000245 \\ 0.000328 & 0.000245 & 0.001333 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0.591 & 0.531 \\ 0.591 & 1 & 0.418 \\ 0.531 & 0.418 & 1 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 & 0 \\ 0.591 & 1 & 0 \\ 0.531 & 0.418 & 1 \end{bmatrix}$

Question 7

Given Cholesky decomposition $A = LL^T$ and defining matrix L as:

$$\begin{bmatrix} 0.75 & 0 \\ -0.05 & 0.46 \end{bmatrix}$$

which of the following corresponds to A?

- $\begin{bmatrix} 0.5625 & -0.0375 \\ -0.0375 & 0.2141 \end{bmatrix}$
- $\begin{bmatrix} 0.5625 & 0 \\ -0.0375 & 0.2141 \end{bmatrix}$
- $\begin{bmatrix} 0.5625 & -0.0375 \\ 0 & 0.2141 \end{bmatrix}$
- $\begin{bmatrix} 0.75 & -0.05 \\ -0.05 & 0.46 \end{bmatrix}$

Question 8

(**Note:** Use Python to solve this question.) Get daily adjusted closing prices data using the yfinance library in Python for the period spanning Jan. 1st, 2015, to Dec. 31st, 2020, for the company Bank of America (Ticker: BA). Calculate the annual volatility of its returns in the period (assume 252 trading days in a year). How much is it?

- 52.45%

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- 3.67%
- 42.52%
- 2.68%

Question 9

(**Note:** Use Python to solve this question.) Use the **yfinance library** to extract daily stock price data for the firm Apple (Ticker: AAPL). Use Adjusted Close prices to calculate discretely compounded returns. What would have been by July 30th, 2021, the value of a $10,000 investment in the company on the first trading day of January 2021?

- $1,307.17
- $11,307.17
- $1,228.52
- $11,228.52

Question 10

(**Note:** Use Python to solve this question.) Use the yfinance library in Python to download daily adjusted closed price data for the tickers CSCO, GE, and MMM. Use the period spanning Jan. 1st to Dec. 31st of 2021. Calculate the correlation matrix (M) of their simple daily returns. What is the Cholesky decomposition matrix (L) such that $LL^T = M$?

- $\begin{bmatrix} 1 & 0.3020 & 0.4493 \\ 0 & 0.9533 & 0.2886 \\ 0 & 0 & 0.8454 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 & 0 \\ 0.3020 & 0.9533 & 0 \\ 0.4493 & 0.2886 & 0.8454 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 & 0 \\ 0.2565 & 0.9533 & 0 \\ 0.3445 & 0.3345 & 0.8454 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 & 0 \\ 0.3020 & 1 & 0 \\ 0.4493 & 0.4109 & 1 \end{bmatrix}$

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Question 11

You are planning to use a GBM process to model prices of two correlated assets (A and B). You have sampled two values from a univariate standard normal such that $x_A = 0.44$ and $x_B = 0.26$. The correlation of asset B with A is 1. Annualized volatility of asset B is 15%. What will be the next value of the volatility term (i.e., σdW_t) of asset B if you are doing one-month steps in the simulation?

- -0.066
- -0.01905
- 0.01905
- 0.066

Question 12

(**Note:** Use Python to solve this question.) Get daily adjusted closing prices using the yfinance library in Python for the period spanning Jan. 1st, 2001, to Dec. 31st, 2008, for the companies Ford (Ticker: F) and Lockheed Martin (Ticker: LMT). Calculate the correlation coefficient of daily returns (you can use the numpy function 'corrcoef') for the period.

- 1
- 0.2283
- 0.0123%
- 0.0123

Question 13

(**Note:** Use Python to solve this question.) Use the **yfinance library** to extract daily stock price data for the firm Ford (Ticker: F) in the period January 1st, 2021, to December 31st, 2021 (available days in the interval). Use Adjusted Close prices to calculate returns. What is the

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QUESTIONS

1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16

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discretely compounded return for that stock in the period?

- 88.15%
- 141.46%
- 151.43%
- 1.52%

Question 14

(**Note:** Use Python to solve this question.) Suppose you want to simulate stock prices for three correlated assets following a GBM process. The correlation matrix of these assets (in order A, B, and C) is:

$$\begin{bmatrix} 1 & 0.772 & -0.023 \\ 0.772 & 1 & 0.351 \\ -0.023 & 0.351 & 1 \end{bmatrix}$$

If you are assuming a GBM process for each asset of the form:

$$dS/S = \mu dt + \sigma dW_t, \quad \text{where} \quad dW_t = \epsilon\sqrt{dt}$$

Use Python to calculate, respectively, the ϵ_i corresponding to each asset A, B, and C, respectively, for the next step of the simulation. Use a random seed of 42 when generating random numbers using np.random.standard_normal(3).

- $\epsilon_A = -0.416$ $\epsilon_B = -0.324$ $\epsilon_C = -2.076$
- $\epsilon_A = 0.496$ $\epsilon_B = -0.138$ $\epsilon_C = 0.647$
- $\epsilon_A = 1.764$ $\epsilon_B = 1.616$ $\epsilon_C = 0.988$
- $\epsilon_A = 0.496$ $\epsilon_B = 0.295$ $\epsilon_C = 0.435$

Question 15

You are planning to use a GBM process to model prices of two correlated assets (A and B). You have sampled two values from a univariate standard normal such that $x_A = 0.44$ and $x_B = -1.26$. The correlation of asset B with A is 0.25.

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Annualized volatility for both assets is 35%. What will be the next value of the volatility term (i.e., σdW_t) of asset B if you are doing one-month steps in the simulation?

- -1.1099
- -0.112
- 1.219
- -0.3204

Question 16

(**Note:** Use Python to solve this question.) Get daily adjusted closing prices using the yfinance library in Python for the period spanning Jan. 1st, 2005, to Dec. 31st, 2010, for the companies Bank of America (Ticker: BA) and JP Morgan (Ticker: JPM). What is the covariance of their daily returns in the period?

- 0.0413%
- 0.000413%
- 0.0298%
- 0.000298%

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